MIL-I-43682D
27 December 1982
SUPERSEDING
MIL-I-43682C
18 December 1978

MILITARY SPECIFICATION

ICEMAKER-DISPENSER: WITH WATER DISPENSER ATTACHMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers automatic icemakers with water dispensers, capable of making and storing ice and dispensing ice and water. The unit can be completely self-contained or have a remote icemaking unit (see 6.1).

1.2 Classification. The dispensing unit shall be of the following types, grades, sizes, and styles as specified (see 6.2):

Type I	-	Air cooled condenser
Type II	-	Water cooled condenser
Grade A	-	Corrosion-resisting steel
Grade B	-	Low carbon steel
Size 1	-	525 pounds per day (minimum at 70°F) ambient air temperature
Size 2	-	200 pounds per day (minimum at 70°F) ambient air temperature
Style 1	-	Floor style (sizes 1 and 2)
Style 2	-	Counter style (sizes 1 and 2)
Style 3	-	Wall style (size 2)
Style 4	-	Remote style (sizes 1 and 2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4110

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390	-	Plastic Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium, and High Density)
BB-F-1421	-	Fluorocarbon Refrigerants
HH-I-524	-	Insulation Board, Thermal (Polystyrene)
QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections
WW-T-775	-	Tube, Copper, Seamless (for Air Conditioning and Refrigeration)

MILITARY

MIL-V-173	-	Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-F-1183	-	Fitting, Tube, Cast Bronze, Silver-Brazing
MIL-F-3541	-	Fittings, Lubrication
MIL-B-11040	-	Belt, V, Engine Accessory Drive
MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing Of
MIL-C-15726	-	Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings
MIL-B-16541	-	Bronze Valve Castings
MIL-F-20042	-	Flanges, Pipe, Bronze, (Silver Brazing)
MIL-B-24059	-	Bronze, Nickel Aluminum, Rod, Flat Products with Finished Edges, Shapes and Forgings

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-889	-	Dissimilar Metals
MS 17291	-	Receivers, Liquid, Refrigerant 12
MS 17797	-	Valve, Regulating, Fluid Pressure, (Water) Refrigerant-12

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:01Z
Check the source to verify that this is the current version before use.

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

RUBBER MANUFACTURER'S ASSOCIATION

Engineering Standards for Multiple V-Belt Drives

(Application for copies should be addressed to the Rubber Manufacturer's Association, 444 Madison Avenue, New York, NY 10022.)

AMERICAN SOCIETY FOR HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS (ASHRAE)

Standard No. 29 - Methods of Rating and Testing Ice Makers (February 1971)

(Application for copies should be addressed to the American Society for Heating, Refrigerating and Air Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL SANITATION FOUNDATION (NSF)

Listing of Food Service Equipment

Standard No. 12 - Automatic Ice Making Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P. O. Box 1468, Ann Arbor, MI 48106.)

AIR-CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

Standard 810 - Standard for Automatic Commercial Icemakers

(Application for copies should be addressed to the Air-Conditioning and Refrigeration Institute, 1815 North Fort Myer Drive, Arlington, VA 22209.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167	-	Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 506	-	Steel Sheet and Strip, Alloy, Hot-Rolled, and Cold Rolled Regular Quality
B 61	-	Steam or Valve Bronze Castings

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:01Z
Check the source to verify that this is the currentversion before use.

B 148 - Aluminum Bronze Sand Castings
B 150 - Aluminum Bronze Rod, Bar, and Shapes
B 584 - Copper Alloy Sand Castings for General Applications

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORIES INC.

Standard No. 563 - Icemaker

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard product. The icemaker-dispensers delivered under this document shall be the manufacturer's standard commercial product except for changes necessary to meet the document requirements. All like items furnished on any one contract, including parts and sub-assemblies thereof, shall be new and interchangeable.

3.2 Codes and standards. The icemaker-dispenser shall comply with the applicable requirements of NSF Standard No. 12 and UL Standard No. 563.

3.2.1 Compliance. Prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative, satisfactory evidence that the icemaker-dispenser he proposes to furnish under this document meets the applicable requirements of NSF Standard No. 12 and UL Standard No. 563 as follows:

3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished icemaker-dispenser, or

(2) A certification for the icemaker issued by NSF under their special one time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the icemaker-dispensers are being procured, indicating that the icemaker-dispensers have been tested and conform to NSF Standard No. 12.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:01Z
Check the source to verify that this is the current version before use.

3.2.1.2 UL. Acceptable evidence of meeting the applicable requirements of UL shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the icemaker-dispensers have been tested and conform to UL Standard No. 563.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial icemaker-dispenser provided the completed item complies with all provisions of this document (see 6.6).

3.3.1 Refrigerant. The refrigerant shall conform to type 12, 22, or 502 of BB-F-1421.

3.3.2 Tube, copper. Copper tube used for refrigerant lines shall conform to type II of WW-T-775.

3.3.3 Steel.

3.3.3.1 Stainless steel. Stainless steel shall conform to ASTM A 167.

3.3.3.2 Carbon. Low carbon steel shall conform to ASTM A 506.

3.3.4 Insulation. Insulation shall be non-moisture absorbing material; shall be homogeneous throughout; and shall have "k" factors specified below, using a 75° Fahrenheit (F) mean temperature and expressed in terms of British thermal units per hour (Btuh) per square foot per inch of thickness per degree F temperature difference. Loose fill type insulation will not be acceptable.

3.3.4.1 Pre-expanded insulation. Pre-expanded insulation shall be polystyrene material conforming to class 2, types I or II of HH-I-524.

3.3.4.2 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial rated "k" factor of not more than 0.15, a core density of not less than 1.6 pounds per cubic foot when tested as specified in 4.3.5.

3.3.4.3 Semirigid insulation. Semirigid insulation shall be "B" fiber or finer, industrial type fiberglass with a manufacturer's commercial rated "k" factor of not more than 0.260, with a density of at least 1.0 but not more than 3.0 pounds per cubic foot (see 4.2.1.1.1) and shall be held together with a thermosetting binder.

3.3.5 Fasteners. Screws, bolts, nuts, and other fasteners shall be corrosion-resisting steel or low carbon steel, zinc or nickel plated. Threads shall conform to FED-STD-H28.

3.3.6 Plastic. The plastic shall conform to type I, class 4, grade 1, of L-P-390.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

3.4 Design. The icemaker-dispenser shall be a completely self-contained unit for styles 1, 2, and 3. The style 4 shall have a remote condensing unit and shall be connected to the dispenser section by a flexible plastic tube conforming to the material specified in 3.3.6. The type I shall have and air cooled condenser and the type II shall have a water cooled condenser. The icemaker-dispenser shall be a floor style for style 1 and shall be furnished in sizes 1 and 2; a counter style for style 2 and shall be furnished in sizes 1 and 2; and a wall style for style 3 and shall be furnished in size 2. The style 4 shall be a remote style and shall be furnished in sizes 1 and 2. When specified for Army (see 6.2), the size 2 shall not exceed the following dimensions, 33 inches long by 34 inches wide by 86 inches high; otherwise all dimensions shall be the manufacturer's standard commercial dimensions. Unless otherwise specified (see 6.2), the icemaker-dispenser shall produce ice shaped either (1) as defined in ASHRAE Standard No. 29; (2) block shaped ice: (3) cracked ice having irregular shape. The icemaker-dispenser shall have an evaporator to which water is applied and frozen to form ice, a water dispensing mechanism, a means of harvesting and storing ice, automatic ice level control, motor or motor drives, and all necessary piping, wiring, control and safety devices and accessories required to provide an automatic operation. Controls shall be designed for operation on nominal 120 volt, 1-phase, 60 hertz (Hz), power source. Transformers are permitted as part of the control system, but any transformers, relays, or other equipment necessary for a complete control system shall be furnished and installed as an integral part of the ice making machine. The ice water shall be dispensed to the user in a sanitary manner. Unless otherwise specified (see 6.2), the icemaker-dispenser shall have a continuous flow. All parts shall be located to permit convenient repair or replacement. Those parts that transmit vibration during normal operation shall be mounted on vibration elimination units. All accessories, including fans, reduction gears, harvesting equipment, and pumps shall be direct connected, V-belt driven or chain driven from a reduction gear. When the icemaker-dispenser is operated, all mechanical and electrical controls and parts shall function according to design (see 4.3.4). When specified for Navy shipboard use (see 6.2), the equipment shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other with no spillage of water or ice when tested as specified in 4.3.7. When specified for Navy shipboard use (see 6.2), all types, sizes, and styles of ice making machines shall be equipped with key lock switches to prevent dispensing of ice. The key lock switch shall not interfere with the machines ice making capabilities. Two keys shall be provided with each dispenser.

3.4.1 Capacity. The icemaker-dispenser shall be capable of making and harvesting ice under the conditions in table I, when tested in accordance with 4.3.2. When specified for Navy shipboard (see 6.2), the icemaker-dispenser shall be capable of making and harvesting ice having a value of not less than 122 BTU per pound, under the conditions in table II when tested as specified in 4.3.2.1.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:01Z
Check the source to verify that this is the current version before use.

TABLE I. Capacity conditions

Type	Size	Ambient air temp. degrees (F)	Potable water supply temp. degrees (F)	Condenser inlet water temp. degrees (F)	Minimum capacity for 60 Hz machines lb/24 hrs	Minimum capacity for 50 Hz machines lb/24 hrs
I	1	70	70	None	525	438
		100	90	None	400	333
I	2	70	70	None	200	167
		100	90	None	140	117
II	1	70	70	70	525	438
		100	90	90	400	333
II	2	70	70	70	200	167
		100	90	90	140	117

TABLE II. Capacity conditions (For Navy Shipboard)

Type	Size	Ambient air temp. degrees (F)	Potable water supply temp. degrees (F)	Condenser inlet water temp. degrees (F)	Minimum capacity lb/24 hrs
I	1	90	70	None	525
I		100	90	None	400
I	2	90	70	None	200
		100	90	None	140
II	1	90	70	70	525
		100	90	90	400
II	2	90	70	70	200
		100	90	90	140

3.5 Construction. The icemaker-dispensers shall be totally inclosed in a structural frame and sheet metal housing with removable panels; shall show no signs of failure after a 3 inch drop when tested in accordance with 4.3.1; shall be equipped with a drain sink with an antisplash apron below a touch fill water and ice station, a condensing unit, evaporator, and a water system. Joints between dissimilar metals which may become wet in service shall be protected against electrolytic or galvanic corrosion by proper selection of materials, coatings, isolating, gasketing, or similar

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

means. Any bolts, nuts, pins, screws, and other fastenings and fittings of metal shall be of the same material as the materials being joined or shall be cathodic to the materials being joined. Dissimilar metals are rated in MIL-STD-889. When specified for Navy shipboard use (see 6.2), the cabinet construction shall provide for fastening the equipment to a deck. The minimum bolt sizes, using four standard bolts, shall be 1/2-inch for equipment not over 400 lbs and 5/8-inch for equipment not over 700 lbs.

3.5.1 Condensing unit. The condensing unit shall be designed to operate on the frequency and voltage specified. The use of any transformer to obtain the specified frequency and voltage is prohibited.

3.5.1.1 Compressor. The compressor shall be open type or hermetic, mechanically or thermally sealed. The refrigerating system shall be fitted with means of discharging or adding refrigerant to the system. When receivers are used, the compressor shall be fitted with suction and discharge valves. The assembled refrigeration system shall be completely dehydrated and charged with the amount of refrigerant and oil necessary for operation.

3.5.1.1.1 Open type compressor. When open type compressors are used they shall be V-belt driven. V-belts shall be static conductive and conform to the Engineering Standards for Multiple V-belt Drives of the Rubber Manufacturer's Association. In addition, the belt or belts shall conform to MIL-B-11040 for electrical conductance. The seal of open type compressors shall be replaceable without the use of special tools and without regrinding the shaft shoulder.

3.5.1.1.2 Compressor motor. Unless otherwise specified (see 6.2), the compressor motors, up to and including 3/4 horsepower shall operate on a nominal 120-volt, 60 hertz, 1-phase power source. Compressor motors over 3/4 horsepower shall operate on a nominal 240-volt, 60 hertz, 1-phase power source. Motors shall be protected against damage from overloads and voltage fluctuations when starting or running by means of a thermal overload protector. When 50 Hz frequency is specified, the motor shall be specifically designed for operation on that frequency.

3.5.1.2 Condenser. The type I icemaker-dispenser condenser shall be air cooled, constructed of copper, aluminum, or steel, or a combination thereof. The type II icemaker-dispenser condenser shall be water cooled of the shell and coil type, the multipass shell and tube type or the tube in tube type with provisions for cleaning the water circuit without opening the refrigerant circuit. The shell shall be steel and tubes and coils shall be copper. The water cooled condenser shall be equipped with a water regulating valve conforming to MS 17797. When type II units are specified for Navy shipboard use (see 6.2), the condenser shall be watercooled and designed for use with sea water at an entering temperature of 85°F and not more than 110°F condensing temperature. Water velocities in the condenser shall meet the condensing capacity with sea water entering the condenser at 90°F. The condenser shall be a type in which the tubes can be mechanically cleaned and shall be installed so that the tubes may be cleaned without removing the condenser from the equipment. All parts in

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

contact with sea water shall be made of copper-nickel (90-10) conforming to MIL-C-15726, aluminum bronze conforming to MIL-B-24059, ASTM B 584 alloy 1 B, ASTM B 148 alloy 9 D, ASTM B 150 alloy 2, or valve bronze conforming to ASTM B 61 except the condenser sea water tubes shall be copper-nickel (90-10) conforming to MIL-C-15726. The heads may be Delrin (AF) plastic or a simlar material satisfactory to NAVSEA (see 6.4). The condenser shall be provided with specific devices to purge both the refrigerant and water circuits and for draining the water circuit. A water-regulating valve shall be provided with each condenser. The valve shall be of the direct-acting or pilot-controlled type actuated by condenser gas pressure to modulate the flow of water required for the condenser. The valve shall be selected to regulate water flow from shut off to required capacity within a maximum operating gas pressure rise of 40 pounds per square inch (psi). The valve shall be adjustable within an operating range of 90 to 140 psi refrigerant gas pressure for R-12, or equivalent, where other refrigerants are used. The capacity of the water-regulating valve shall at least equal the water requirement of the condenser, based on available inlet water pressure of 35 psi and a pressure drop across the valve of not more than 10 psi. The valve body shall be made of valve bronze conforming to MIL-B-16541. In addition, the internal metal parts subject to corrosion or erosion shall be made of nickel-copper conforming to QQ-N-281. The valve shall have either flanged connections conforming to MIL-F-20042 or union ends conforming to MIL-F-1183. The strainer body shall be valve bronze of ASTM B 61, (90-10) copper-nickel, monel or other materials satisfactory to the using or procuring activity. The valve shall be constructed in such a manner as to prevent the possibility of entry of sea water to the refrigerant system in the event of derangement. A strainer of No. 20 mesh nickel-copper screen of substantial area shall be provided for the protection of the valve. The strainer basket shall be readily removable for cleaning without disassembly of connecting piping. The strainer shall be not less than 3/4-inch nominal.

3.5.1.2.1 Receivers. Receivers, if used, shall conform to MS 17291. The receivers shall be fitted with a back seating outlet valve, a 1/8 inch National Pipe Thread (NPT) opening, closed with a fusible plug. Combination water cooled receiver condensers will be acceptable for type II icemaker-dispensers.

3.5.2 Evaporator. Parts of the evaporator contacting ice shall be brass, stainless steel (see 3.3.3.1), anodized aluminum or aluminum alloy, or other non-toxic material having resistance to corrosion by water and ice.

3.5.3 Cutting and harvesting blades. Cutting and harvesting blades and components shall be fabricated of stainless steel (see 3.3.3.1).

3.5.4 Water circulator or pump. The water circulator or pump, when used, shall have a capacity adequate to provide a continuous and equal volume of water over the entire ice making surface. Pumps shall be provided with a brass drain plug or shall be of self-draining design.

3.5.5 Water tank. Water tanks, when used, shall be corrosion-resisting steel (see 3.3.3.1) or other corrosion-resistant material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

3.5.6 Dehydrator. A dehydrator shall be installed in each condensing unit.

3.5.7 Tube. Tube shall be as specified in 3.3.2. All tube connections shall be solder except flared connections shall be provided on the expansion valve inlet and outlet connections, receiver outlet valve, and other locations normal to the manufacturer's standard product. When specified for Navy shipboard use (see 6.2), all piping shall be securely supported to minimize strain and vibration. Sea water piping shall be (90-10) copper-nickel. Fittings in the sea water piping shall be (90-10) copper-nickel, valve bronze ASTM B 61 or monel. Welded fittings may be used. Electrical switches, terminal blocks or wiring connections shall be protected from cold piping.

3.5.8 Cabinet. The cabinet for the grade A icemaker-dispenser shall be fabricated of the material specified in 3.3.3.1, no thinner than 24 gage (0.025 nominal). The grade B icemaker-dispenser cabinet shall be fabricated of the material specified in 3.3.3.2, no thinner than 22 gage (0.029 nominal). The cabinet frame shall be constructed of stainless steel or aluminum angle not less than 1 inch by 1 inch by 3/32 inch or of structural members having the same or better section modulus, welded or bolted into an integral unit.

3.5.8.1 Mounting. The cabinet shall be mounted on legs or shall be designed for securing directly to the floor or counter, as applicable. When specified for Navy shipboard use (see 6.2) style 1 icemaker-dispensers shall be provided with four removable legs suitable for bolting the ice-maker-dispenser to the deck. Legs shall be 6 inches long, fabricated from 14 gauge minimum stainless steel (3.3.3.1) and shall be adequately reinforced. The bottom flange of each leg shall be provided with 9/16 inch diameter cleanance hole for hold down bolt. Each hold down bolt shall be accessible from the front for mounting and dismounting the icemaker-dispenser. Legs shall attach to a structural member or reinforcing plate on the cabinet base of the icemaker-dispenser. Attaching holes in the bottom of the icemaker-dispenser base may be tapped (minimum depth one bolt diameter plus one thread) or captive nuts may be utilized. Style 2 icemaker-dispensers shall be provided with 4 inch long stainless steel, solid, round stock legs 1 inch minimum diameter with each end drilled and tapped 5/8 inch minimum depth for attaching bolts.

3.5.9 Storage reservoir. The minimum capacity of the ice storage reservoir shall be as specified by the contracting officer (see 6.2). For the size 1 icemaker it shall be either 95, 45, or 25 pounds. For the size 2 icemaker it shall be either 30 or 8 pounds. Ice storage capacity shall be tested as specified in 4.3.3. The ice storage reservoir shall be designed to permit the removal of all the ice through the ice dispensing mechanism when tested as specified in 4.3.3. The ice storage reservoir shall be lined with corrosion-resisting steel specified in 3.3.3.1 or plastic specified in 3.3.6 when foamed-in-place insulation is used. The thickness of the liner shall be 0.025 inch, except when foamed-in-place insulation is used the thickness of the liner shall be 0.0125 inch. All joints and corners shall be welded or soldered and ground smooth with all inside seams filled. The ice storage reservoir shall have a drainage opening.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:01Z
Check the source to verify that this is the currentversion before use.

3.5.10 Insulation. The ice storage reservoir for the icemaker-dispensers shall be insulated with material specified in 3.3.4 The insulation thickness shall be determined by the manufacturer so that the computed overall heat transfer coefficient, "U", is not more than 0.45 Btuh per square foot for the sides of the ice storage reservoir. The computed "U" value shall be the "k" of the insulation value divided by the thickness of the insulation used. Rigid insulation, preformed or foamed-in-place shall be used for the bottom ice storage reservoir. When a semi-rigid insulation is used for the top and four sides, it shall be cut slightly larger and thicker than the shape and space to be filled, so as to require forcing into place.

3.5.11 Fungus resistance. When specified (see 6.2), electrical connections, including terminal and circuit connections, components and circuit elements shall be coated with varnish conforming to MIL-V-173 except that:

(1) Components and elements inherently inert to fungi or hermetically sealed need not be treated.

(2) Components and elements whose operation will be adversely affected by the application of varnish shall not be treated.

3.5.12 Lubrication. All moving parts and other surfaces requiring lubrication shall be provided with accessible means for applying the required lubricants. Fan shaft bearings shall be lubricated from one general location and shall be clearly identified. The equipment shall be designed to operate efficiently and satisfactorily when lubricated with standard Military lubricants (see 6.3). Grease fittings shall conform to MIL-F-3541.

3.5.13 Caution plates. Where the use of high pressure lubricating equipment, 1,000 pounds per square inch gage (psig) or higher, will damage greases seals suitable warning shall be affixed to the equipment in a conspicuous location.

3.5.14 Initial lubrication service. Unless otherwise specified (see 6.2), equipment shall be serviced with winter grades of standard Army or Navy lubricants designated for use in the temperature range of 0°F to plus 42°F. After the lubrication services, each item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grades of lubricants used. Points requiring lubrication shall be accessible and clearly identified by red painted markings surrounding each lubricating fitting.

3.6 Finish. Unless otherwise specified (see 6.2), the exterior shall have the manufacturer's standard commercial finish, provided it conforms to NSF requirements. All burrs, rough edges, and slivers shall be removed.

3.7 Electromagnetic compatability. When specified (see 6.2), the icemaker-dispenser unit shall comply with the test limits for conducted (CE04 and CE05) and radiated emission limits (RE02) in accordance with part 10, class C3, group I of MIL-STD-461 (see 4.3.6). The icemaker-dispenser shall be exempt from Short Duration Interference test requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

3.8 Identification plate. The identification plate shall be the manufacturer's standard plate providing it is weather-resistant and will not scratch or fade. The icemaker-dispenser shall be marked in accordance with MIL-STD-130 and with the following:

(a) Nomenclature
(b) Type of unit, capacity of unit, and type of refrigerant and charge
(c) National stock number
(d) Contract and technical manual number
(e) Military model number
(f) Manufacturer's name, model, and serial number
(g) Manufacturer's address
(h) U.S.

For Navy shipboard use (see 6.2), the identification plate shall be located so that it is readily readable from the front of the icemaker-dispenser.

3.9 Instructions. Instructions for start-up, operating, defrosting, shut-down, cleaning and sanitizing and preventative maintenance procedures shall be displayed in letters not less than 1/16 inch high. Wiring diagrams and refrigeration cycle diagrams shall also be displayed. The manufacturer's standard method of display may be used provided the legends are protected against weathering and abrasion. All instructions shall be located in accordance with the manufacturer's commercial practice.

3.10 Servicing. For Navy shipboard use (see 6.2), icemaker-dispensers shall be constructed so that all parts requiring service are readily accessible and normal servicing can be effected from the front of the icemaker-dispenser.

3.11 Workmanship. The finished icemaker-dispensers shall be clean and all forming and cutting shall be done accurately and cleanly. All scale and flux deposits shall be removed from the finished welds. All exposed welds shall be ground smooth to match the adjacent surfaces. All nuts, bolts, and screws shall be tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202640-21T22:01Z
Check the source to verify that this is the currentversion before use.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.1.1 Insulation density test.

4.2.1.1.1 Foamed-in-place insulation. The density test for foamed-in-place insulation shall be performed as specified in 4.3.5. The inspection lot shall be the amount of material used in one day's production. The sample unit shall be a 6 inch cube.

4.2.1.1.2 Semi-rigid insulation. The density of semi-rigid insulation shall be tested for compliance with 3.3.4.3 by weighing a sample quantity consisting of at least five cubic feet. The inspection lot shall be the amount of insulation offered for inspection at one time.

4.2.2 Certification. Certified test reports and/or UL labels shall be examined for proof of compliance with the requirements of 3.2.1.

4.2.3 In-process inspection. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

(a) Insulation is installed in accordance with 3.5.10.
(b) Joints between dissimilar metals are as specified in 3.5.
(c) Lubrication is in accordance with 3.5.14.
(d) Varnish coating when required by 3.5.11.

4.2.4 End item visual examination. The end item shall be examined for the defects in table III. The lot size shall be expressed in units of icemaker-dispensers. The sample unit shall be one icemaker-dispenser. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

TABLE III. Visual defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's commercial finish		X
	Burrs, rough edges, or slivers	X	
Design, construction and workmanship general (applicable to all components and assemblies)	Any component missing, inoperative, or not type specified	X	
	Any component fractured, split, punctured, torn, dented, creased, loose, bowed, or sprung	X	
	Forming and cutting not accurate or cleanly performed		X
	Water valve not adjustable as required		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

TABLE III. Visual defects (cont'd)

Examine	Defects	Classification Major	Minor
Bolts, nuts, screws	Loose, missing		X
Caution plates	Caution plates missing or not as specified when applicable	X	
Wiring	Wiring diagram missing		X
Welding	Scale or flux deposits not removed	X	
	Welds not ground smooth	X	
Lubrication	Lubrication tags missing or not as specified		X
	Lubrication fittings not painted red		X
Marking for identification	Missing, incomplete, not legible, not specified type, not affixed or marked in prescribed manner, or not of specified material		X
Instructions	Incorrect, missing, not legible, not protected	X	
	Misplaced, not as specified		X

4.2.5 Dimensional examination. The icemaker-dispenser shall be examined for compliance with dimensions specified. Any dimension not within specified requirements shall be classified a defect. The inspection level shall be S-2 and the AQL expressed in terms of defects per hundred units shall be 4.0.

4.2.6 End item testing.

4.2.6.1 Operational testing. Each icemaker-dispenser shall be tested in accordance with 4.3.4. Any test failure shall be cause for rejection of the icemaker-dispenser failing the test.

4.2.6.2 Performance testing. One representative unit from the first production lot shall be tested as specified in 4.3.1, 4.3.2, 4.3.3, 4.3.6 and 4.3.7. Any nonconformance shall be cause for rejection of the lot. If the representative unit passes all of the tests in 4.3.1, 4.3.2, 4.3.3, 4.3.6 and 4.3.7, subsequent lots produced without design, material, or manufacturing changes may be accepted without repetition of those tests.

4.2.7 Packaging inspection. An examination shall be made to determine that preservation-packaging, packing and marking comply with section 5 requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

4.3 Methods of inspection.

4.3.1 Drop test. The icemaker-dispenser shall be raised once on each side and dropped onto a concrete floor from a height of 3 inches and then operated to determine compliance with the requirements of 3.5. Inability of the unit to operate, refrigerant leaks, or inability of controls to function properly, shall constitute failure of this test.

4.3.2 Capacity test. The icemaker-dispenser shall be tested as follows: After 1/2 hour of operation of the icemaker-dispenser, all of the ice shall be removed from the storage bin and discarded. Then the icemaking machine shall be operated for 4 hours, in accordance with the conditions specified in table I, dispensing the ice every 15 minutes into a perforated container. The weight of ice shall be record-ed. At the end of 4 hours the ice making capacity shall be determined. Inability of the icemaker-dispenser to meet the capacity requirements of 3.4.1 shall constitute failure of this test.

4.3.2.1 Capacity test (for Navy shipboard). The icemaker-dispenser shall be tested in accordance with the methods and conditions prescribed in ARI Standard No. 810, ASHRAE Standard No. 29, and the conditions specified in 3.4.1. Inability of the unit to meet the capacity requirements of 3.4.1 shall constitute failure of this test.

4.3.3 Storage reservoir test. The ice storage reservoir test shall be performed after the capacity test in accordance with the conditions specified in table I or II as applicable. The ice making machine shall be operated until the ice storage reservoir is full and the machine has cycled off on the storage reservoir control. The icemaking machine shall remain on the same ambient conditions for 12 hours without dispensing any ice. After the 12 hours the ice shall be dispensed continuously. During the dispensing the icemaking machine shall not replenish the storage reservoir. When the icemaking-dispensing machine ceases to dispense ice, the interior of the storage reservoir shall be examined to determine that all the ice has been removed in accordance with 3.5.9. The ice shall be weighed to determine compliance with the storage reservoir capacity requirements of 3.5.9. Failure to meet the storage reservoir capacity requirement or inability of the ice dispenser to dispense the ice completely shall constitute failure of this test.

4.3.4 Operational test. The icemaker-dispenser shall be operated through one ice-making and harvesting cycle to determine compliance with 3.4. The icemaker-dispenser shall be tested using electrical characteristics cited in the procurement document. Inability of the unit to operate through one complete cycle and dispense ice and water as specified shall constitute failure of this test.

4.3.5 Density test for insulation. Prior to each day's production one free foam sample of the insulation specified in 3.3.4.2, a minimum size 8 inch cube shall be taken. From the center of the sample a 6 inch cube shall be cut and weighed to the nearest ounce. This weight of 1/8 of a cubic foot, shall be multiplied by 8 to determine conformance to the density requirements. In addition, a daily record of the foam shall be kept as follows, and made available to the contracting officer:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

Calibration of mix (mix proportions)
Amount of flow (per box)
Time of flow (per box)
Temperature of mold

Inability of the 6 inch cube sample to meet the density requirements or failure to keep the above data shall constitute failure of this test.

4.3.6 Electromagnetic interference test. When electromagnetic compatability is required, the first item produced shall be tested by the contractor in accordance with test methods CE04, CE05 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, and the Electromagnetic Compatability Test Report meet the requirements of MIL-STD-462 (see 6.2).

4.3.7 Roll, list, and trim, test. The icemaker-dispenser shall be bolted to a table or platform. The unit shall be started and the table or platform inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other and operated for 1-hour in each plane. During and after both 1-hour periods, the unit shall be examined to determine compliance with 3.4. Failure of the unit to continue operating or any leaks or malfunction of any component or any spillage shall constitute failure of this test.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Each complete unit shall be preserved, packaged, packed and marked in accordance with the applicable requirements of MIL-R-12323. Preservation and packaging shall be level A or C, and packing shall be level A, B, or C as specified, except that each complete icemaker-dispenser shall be shipped in one shipping container (see 6.2).

6. NOTES

6.1 Intended use. The icemaker-dispensers are intended to be used in garrison dining halls, hospitals; and other Military installations including ships where a continuous supply of dispensed ice and water is needed. The size 1 is for capacity feeding ranging from 700 to 1100 persons and the size 2 from 80 up to 140 persons.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) Type, grade, size, and style desired (see 1.2).
(c) When ice shape is different (see 3.4).
(d) When special dimensions are required for Army (see 3.4).
(e) When continuous flow is not required (see 3.4).
(f) When construction is for Navy shipboard use (see 3.4 and 3.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

(g) When key lock switches are required (see 3.4).
(h) When capacity conditions are for Navy shipboard use (see 3.4.1 and 4.3.2.1).
(i) Electrical characteristics of compressor motor, if other than specified (see 3.5.1.1.2).
(j) When condenser is for Navy shipboard use (see 3.5.1.2).
(k) When tubing is for Navy shipboard use (see 3.5.7).
(l) When mounting is for Navy shipboard use (see 3.5.8.1).
(m) Capacity of ice storage reservoir (see 3.5.9).
(n) If fungus treatment is required (see 3.5.11).
(o) If initial lubrication service is waived (see 3.5.14).
(p) If finish is other than specified (see 3.6).
(q) When electromagnetic compatibility is required (see 3.7).
(r) When identification plate is for Navy shipboard use (see 3.8).
(s) When access for servicing aboard Navy ships must be provided for (see 3.10).
(t) Level of preservation, packaging and packing is required (see 5.1).

6.3 Standard lubricants. When requested by the contractor, a list of standard Mililtary lubricants applicable to the equipment covered by this document should be furnished.

(a) For Army use, this information is contained in Department Supply Bulletin SB 38-5-3, "List of Standard Lubricants, Hydraulic Fluids, Liquid Fuels and Preservative Materials."

(b) For Navy shipboard use, this information is contained in appendix A. Lubricants used by the Navy Department of Bureau of Ships pamphlet NBS 431, "Lubricants, General Information; Requirements, and Test Methods Relating to Qualification of Mililtary Symbol Oils."

6.4 Engineering design conference. The following information should be included in the invitation for bid: Within 10 days after award of contract and prior to any work on the preproduction model including ordering materials, the contractor may request the contracting officer, in writing, to arrange an engineering design conference to discuss any questions concerning details of Government drawings, instructions, and technical requirements of this document. Questions for NAVSEA should be addressed to the contracting officer. This conference is to assist the contractor in interpretation of the document, and agreements made shall neither obligate the Government to accept equipment that is not in accordance with this document nor relieve the contractor of his obligation to supply the equipment specified.

6.5 Contract data requirements. Any data items to be delivered under any contract for items covered by this document should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.7 Changes from previous issue. Asterisks are not used in this revision to identify the changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD

Review activities:

Army - MD
Navy - MS, SA
DLA - GS

User activity:

Navy - MC

Preparing activity:

Army - GL

Project No. 4110-0365

☆ U.S. GOVERNMENT PRINTING OFFICE: 1983—605-034/555

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:01Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-I-43682D Icemaker-Dispenser: With Water Dispenser Attachment

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:01Z
Check the source to verify that this is the current version before use.

OLD

EPARTMENT OF THE ARMY

OFFICIAL BUSINESS
ENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Laboratories
ATTN: DRDNA-ES
Natick, MA 01760

OLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:01Z
Check the source to verify that this is the current version before use.